

# Engineering Project Report

Project 056: ESP32 Bluetooth Low Energy (BLE) Beacon Sensor Node

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## Abstract

This technical report presents the comprehensive design, circuit schematics, bill of materials, mathematical modeling, assembly methodology, and experimental testing for **ESP32 Bluetooth Low Energy (BLE) Beacon Sensor Node**. Classified under the **Intermediate (College Level)** tier in the **Wireless IoT Networks** category, this design provides optimized performance, high reliability, and clear practical utility for school and engineering applications.

## Contents

## 1 Introduction & System Overview

The objective of this project is to implement a robust electronic system for **ESP32 Bluetooth Low Energy (BLE) Beacon Sensor Node**. This document details the step-by-step engineering design, component functions, mathematical principles, and experimental results.

## 2 Bill of Materials (Components List)

Table 1: Bill of Materials for ESP32 Bluetooth Low Energy (BLE) Beacon Sensor Node

| S.No. | Component Name           | Qty   | Circuit Function                        |
|-------|--------------------------|-------|-----------------------------------------|
| 1     | Microcontroller / Op-Amp | 1     | Arduino UNO / ESP8266 / ESP32 / LM358   |
| 2     | Transducers / Sensors    | 2     | Interface sensor modules                |
| 3     | LCD / OLED Display       | 1     | Visual data display (16x2 / I2C)        |
| 4     | Relays / Driver MOSFETs  | 2     | Actuator control interface              |
| 5     | Passives and Connectors  | 1 Set | Resistors, capacitors, breadboard / PCB |
| 6     | Power Module             | 1     | 5V / 12V regulated DC source            |

## 3 System Architecture & Methodology

The ESP32 Bluetooth Low Energy (BLE) Beacon Sensor Node integrates hardware sensors and microcontrollers/op-amps. Sensor signals are processed through custom firmware or analog conditioning stages and displayed locally or streamed over Wi-Fi/Bluetooth.

## 4 Circuit Assembly & Testing Procedure

1. Place core active components and microcontrollers/ICs on the breadboard or PCB.
2. Wire DC supply rails (+VCC and GND) with appropriate decoupling capacitors.
3. Connect sensor networks and input signal conditioning circuits.
4. Interface output switches, MOSFET drivers, or visual/acoustic indicators.
5. Apply power and measure operating node voltages to verify proper performance.

## 5 Mathematical Derivations & Governing Equations

$$V_{\text{ADC}} = \frac{\text{ADC\_Value}}{1023} \cdot 5.0 \text{ V} \quad (1)$$

$$\text{Distance} = \frac{\text{Time} \times 0.0343 \text{ cm/s}}{2} \quad (2)$$

## 6 Applications & Engineering Conclusion

Engineering laboratory projects, IoT sensor nodes, automated environmental monitoring, and embedded wireless iot networks.